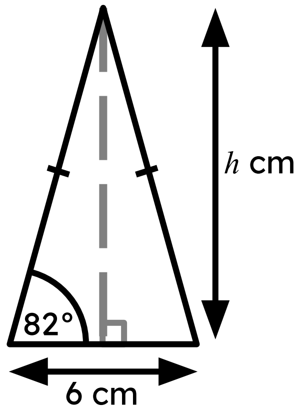
Problem solving with trigonometry

Task A: Calculating areas

1) Calculate the area of each triangle, to 1 decimal place.

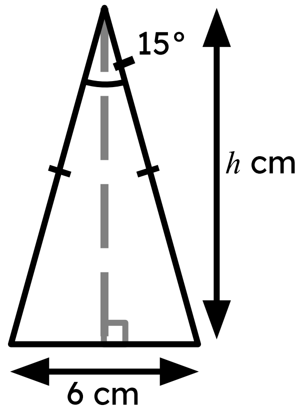
b)



$h = 3 \times \tan(82°) = 21.346\ldots$

$\text{Area} = \frac{1}{2} \times 6 \times 21.346\ldots = 64.0 \text{ cm}^2$

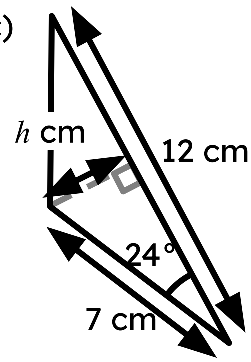
b)



$h = 3 \div \tan(15°) = 11.196\ldots$

$\text{Area} = \frac{1}{2} \times 6 \times 11.196\ldots = 33.6 \text{ cm}^2$

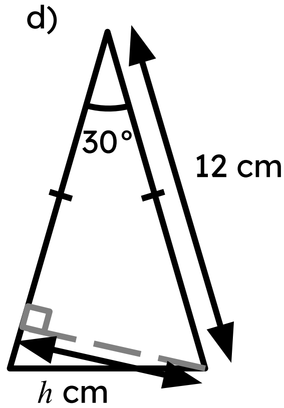
c)



$h = 7 \times \sin(24°) = 2.847\ldots$

$\text{Area} = \frac{1}{2} \times 12 \times 2.847\ldots = 17.1 \text{ cm}^2$

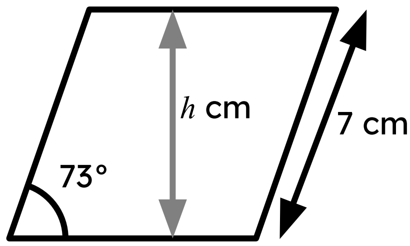
d)

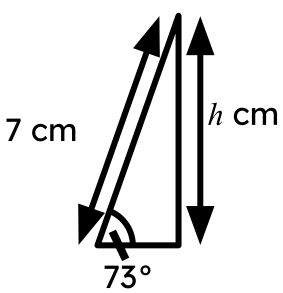


$h = 12 \times \sin(30°) = 6$

$\text{Area} = \frac{1}{2} \times 12 \times 6 = 36 \text{ cm}^2$

2) Calculate the area of this rhombus, to 1 decimal place.

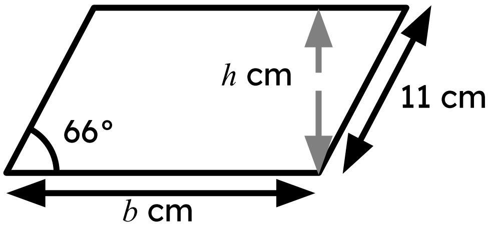




$h = 7 \times \sin(73°) = 6.694\ldots$

$\text{Area} = 7 \times 6.694\ldots = 46.9 \text{ cm}^2$

3) Calculate the area of this parallelogram, to 1 decimal place, given that its perimeter is 54 cm.



Perimeter: $2b + 22 = 54$

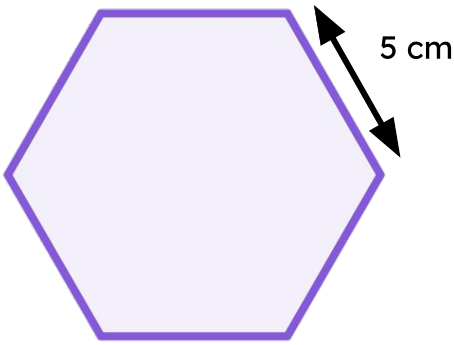
$2b = 32$

$b = 16$

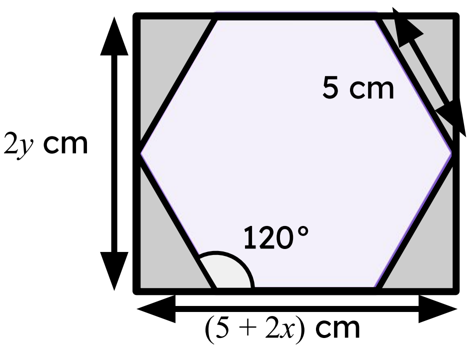
$h = 11 \times \sin(66°) = 10.049\ldots$

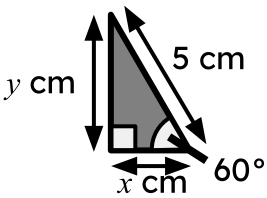
Area = $16 \times 10.049\ldots = 160.8\ \text{cm}^2$

4) Calculate the area of this regular hexagon, to 1 decimal place.



A possible method:





$x = 5 \times \cos(60°) = 2.5$

$y = 5 \times \sin(60°) = 4.33\ldots$

Area of rectangle

$= (2 \times 4.33\ldots) \times (5 + 2 \times 2.5) = 86.602\ldots\ \text{cm}^2$

Area of 4 congruent triangles

$= 4 \times \frac{1}{2} \times 2.5 \times 4.33\ldots = 21.650\ldots\ \text{cm}^2$

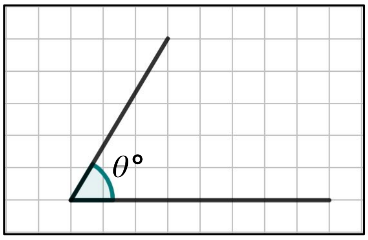
Area of hexagon

$= 86.602\ldots - 21.650 = 65.0\ \text{cm}^2$

Task B: Calculating angles

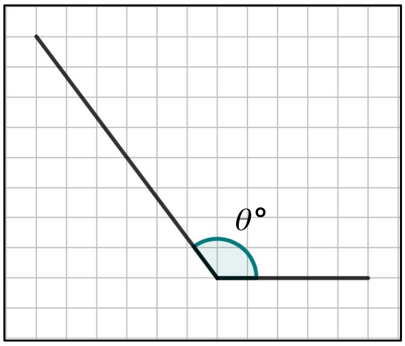
1) Calculate the marked angles on each diagram.

a)



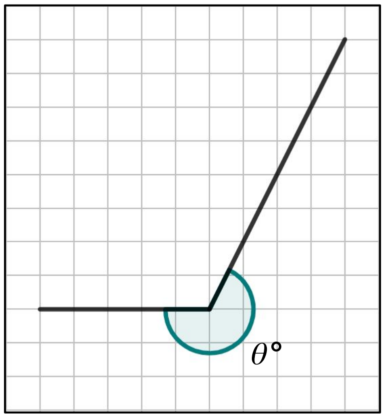
$\theta° = 59.0°$ (1 d.p.)

b)



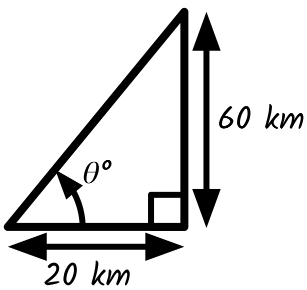
$\theta° = 126.9°$ (1 d.p.)

c)



$\theta° = 243.4°$ (1 d.p.)

2) A yacht is given the instruction to sail 20 km due East and then 60 km due North. If the yacht is already facing East, what angle should it turn to be facing directly towards its final position?

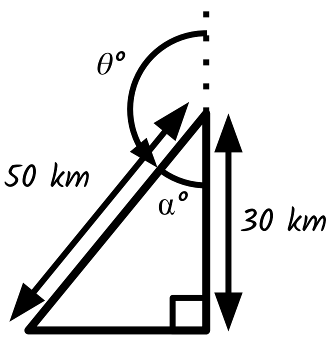


$$\tan(\theta°) = \frac{60}{20}$$

$$\theta° = \tan^{-1}(3)$$

$$\theta° = 71.6° \text{ (1 d.p.)}$$

3) A helicopter travels 30 km due North and then rotates $\theta°$, before travelling 50 km in that direction. The helicopter finishes directly due West of the starting point. What angle of turn did they make?



$$\cos(\alpha°) = \frac{30}{50}$$

$$\alpha° = \cos^{-1}(0.6)$$

$$\alpha° = 53.13\ldots°$$

$$\theta° = 180 - \alpha = 180 - 53.13\ldots = 126.9°$$